Sarthak Hazra

Fifth-Year BS-MS Student

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Education

2022–2027 **Bachelor of Science and Master of Science (BS-MS)**, *Indian Institute of Science Education and Research Tirupati*, Department of Mathematics
CGPA: 8.5/10 | Mathematics CGPA: 9.2/10
Mathematics specialization from the 4th semester, following a common interdisciplinary curriculum.

MS Thesis

June 2026 – Present **Degrees of Loci in Projective Constructions and their Degenerations**, *Advisor: Dr. Chetan Tukaram Balwe*

- **Objective:** Investigate the algebraic mechanisms underlying degree-drop phenomena, and develop a framework for tracking the degrees of loci produced by sequences of projective constructions.
- **Reference:** W. Fulton, *Intersection Theory*.

Academic Interests

Algebraic Geometry, Intersection Theory

Relevant Courses

Analysis	Real Analysis I & II, Complex Analysis, Measure Theory and Integration, Analysis on Euclidean Space, Functional Analysis, ODE
Algebra	Abstract Algebra, Rings & Modules, Fields and Galois Theory, Commutative Algebra, Classical Groups, Representation Theory of Finite Groups
Topology & Geometry	Topology, Algebraic Topology I & II, Differentiable Manifolds and Lie Groups
Other	Elementary Number Theory

Reading Projects

Jan–Apr 2026 **Semester Project**, *Introduction to Schemes*

- **Key Topics:** Sheaves and morphisms of sheaves; construction of $\operatorname{Spec} A$ with its Zariski topology and structure sheaf; schemes and morphisms of schemes; sheaves of modules; divisors.
- **Reference:** R. Hartshorne, *Algebraic Geometry*, Chapter II.

Advisor: *Prof. D. S. Nagaraj*

Aug–Dec 2025 **Semester Project**, *Algebraic Geometry*

- **Key Topics:** Algebraic sets, Hilbert's Nullstellensatz, Zariski topology, regular functions and regular maps, dimension theory, affine varieties as ringed spaces, tangent spaces, tangent cones, singular and nonsingular points.
- **Reference:** J. S. Milne, *Algebraic Geometry* (lecture notes).

Advisor: *Dr. Girja Shanker Tripathi*

May–Jul 2025 **Summer Project**, *De Rham Cohomology and Characteristic Classes*

- **Key Topics:** Exterior algebra, differential forms, chain complexes, de Rham cohomology, Mayer–Vietoris sequences, homotopy invariance, integration on smooth manifolds.
- **Reference:** I. Madsen and J. Tornehave, *From Calculus to Cohomology*.

Advisor: *Dr. Subhash B*

Dec 2024–Apr 2025 **Semester Reading Project**, *Representation Theory of Finite Groups*

- **Key Topics:** Group representations, Maschke's theorem, modules over group algebras, Schur's lemma, irreducible decomposition, character theory and orthogonality relations, induced representations, tensor products.
- **References:** J.-P. Serre, *Linear Representations of Finite Groups*; lectures by Prof. D. S. Nagaraj.

Advisor: *Prof. D. S. Nagaraj*

Talks

15 Apr 2026 **Classification Theorem of Principal G-Bundles** (notes)

A Proof of Hilbert's Nullstellensatz

Extracurricular Activities

Member of the IISER Tirupati Math Club; former member of its core team

Languages

Bengali	Native
English	Professional working proficiency
Hindi	Professional working proficiency

References

1 **Prof. D. S. Nagaraj**
Visiting Professor, Department of Mathematics
Indian Institute of Science Education and Research Tirupati
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2 **Dr. Girja Shanker Tripathi**
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